

UNIT-I

(Data Analytics)

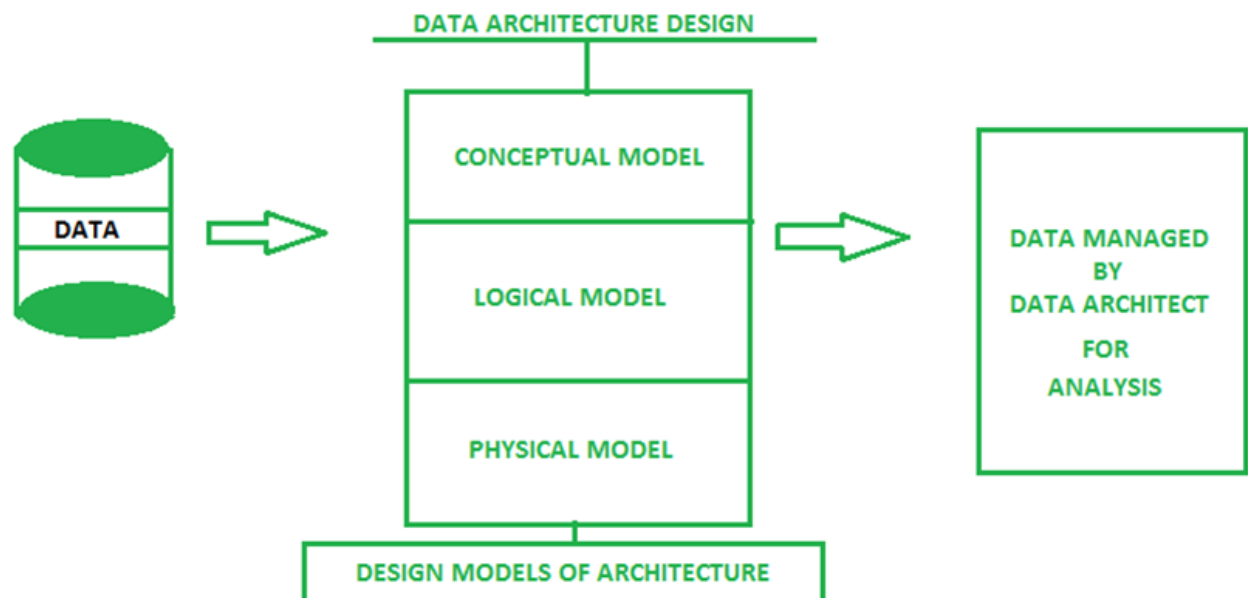
Data Management: Design Data Architecture and manage the data for analysis, understand various sources of Data like Sensors/Signals/GPS etc. Data Management, Data Quality (noise, outliers, missing values, duplicate data) and Data Processing & Processing.

Design Data Architecture:

Data architecture is the process of standardizing how organizations collect, store, transform, distribute, and use data. The goal is to deliver relevant data to people who need it, when they need it, and help them make sense of it. Data architecture design is set of standards which are composed of certain policies, rules and models.

Data is usually one of several architecture domains that form the pillars of an enterprise architecture or solution architecture. The data architecture is formed by dividing into three essential models

- Conceptual model
- Logical model
- Physical model



Conceptual model:

It is a business model which uses Entity Relationship (ER) model for relation between entities and their attributes.

Logical model: It is a model where problems are represented in the form of logic such as rows and column of data, classes, xml tags and other DBMS techniques.

Physical model:

Physical models hold the database design like which type of database technology will be suitable for architecture.

Factors that influence Data Architecture:

Few influences that can have an effect on data architecture are business policies, business requirements, Technology used, economics, and data processing needs.

- Business requirements
- Business policies
- Technology in use
- Business economics
- Data processing needs

Business requirements:

These include factors such as the expansion of business, the performance of the system access, data management, transaction management, making use of raw data by converting them into image files and records, and then storing in data warehouses. Data warehouses are the main aspects of storing transactions in business.

Business policies:

The policies are rules that are useful for describing the way of processing data. These policies are made by internal organizational bodies and other government agencies.

Technology in use:

This includes using the example of previously completed data architecture design and also using existing licensed software purchases, database technology.

Business economics:

The economical factors such as business growth and loss, interest rates, loans, condition of the market, and the overall cost will also have an effect on design architecture.

Data processing needs:

These include factors such as mining of the data, large continuous transactions, database management, and other data preprocessing needs.

Data management:

Data management is an administrative process that includes acquiring, validating, storing, protecting, and processing required data to ensure the accessibility, reliability, and timeliness of the data for its users.

Data management software is essential, as we are creating and consuming data at unprecedented rates.

Data management is the practice of managing data as a valuable resource to unlock its potential for an organization. Managing data effectively requires having a data strategy and reliable methods to access, integrate, cleanse, govern, store and prepare data for analytics. In our digital world, data pours into organizations from many sources – operational and transactional systems, scanners, sensors, smart devices, social media, video and text. But the value of data is not based on its source, quality or format. Its value depends on what you do with it.

Motivation/Importance of Data management:

- Data management plays a significant role in an organization's ability to generate revenue, control costs.
- Data management helps organizations to mitigate risks.
- It enables decision making in organizations.

What are the benefits of good data management?

- Optimum data quality
- Improved user confidence
- Efficient and timely access to data
- Improves decision making in an organization

Managing data Resources:

- An information system provides users with timely, accurate, and relevant information.
- The information is stored in computer files. When files are properly arranged and maintained, users can easily access and retrieve the information when they need.
- If the files are not properly managed, they can lead to chaos in information processing.
- Even if the hardware and software are excellent, the information system can be very inefficient because of poor file management.

Areas of Data Management:

Data Modeling: Is first creating a structure for the data that you collect and use and then organizing this data in a way that is easily accessible and efficient to store and pull the data for reports and analysis.

Data warehousing: is storing data effectively so that it can be accessed and used efficiently in future

Data Movement: is the ability to move data from one place to another. For instance, data needs to be moved from where it is collected to a database and then to an end user.

Understand various sources of the Data:

Data are the special type of information generally obtained through observations, surveys, inquiries, or are generated as a result of human activity. Methods of data collection are essential for anyone who wish to collect data.

Data collection is a fundamental aspect and as a result, there are different methods of collecting data which when used on one particular set will result in different kinds of data.

Collection of data refers to a purpose gathering of information and relevant to the subject-matter of the study from the units under investigation. The method of collection of data mainly depends upon the nature, purpose and the scope of inquiry on one hand and availability of resources, and the time to the other. Data can be generated from two types of sources namely

1. Primary sources of data

2. Secondary sources of data

Points	Primary data	Secondary data
Meaning	Data collected by researcher himself	Data collected by other people
Originality	Original or unique information	Not original or unique information
Adjustment	Does not need adjustment, is focused	Needs adjustment to suit actual aim
Sources	Surveys, observations, experiments	Internal records, govt. published data etc.
Type of data	Qualitative data	Quantitative data
Methods	Observation, experiment, interview	Desk research method, searching online etc.
Reliability	More reliable	Less reliable
Time consumed	More time consuming	Less time consuming
Need of investigators	Needs team of trained investigators	Does not need a team of investigators
Cost-effectiveness	Costly	Economical

1. Primary sources of data:

Primary data refers to the first hand data gathered by the researcher himself. Sources of primary data are **surveys, observations, Experimental Methods.**

Survey: Survey method is one of the primary sources of data which is used to collect quantitative information about an items in a population. Surveys are used in different areas for collecting the data even in public and private sectors. A survey may be conducted in the field by the researcher. The respondents are contacted by the research person personally, telephonically or through mail. This method takes a lot of time, efforts and money but the data collected are of high accuracy, current and relevant to the topic.

When the questions are administered by a researcher, the survey is called a structured interview or a researcher-administered survey.

Observations: Observation as one of the primary sources of data. Observation is a technique for obtaining information involves measuring variables or gathering of data necessary for measuring the variable under investigation.

Observation is defined as accurate watching and noting of phenomena as they occur in nature with regards to cause and effect relation.

Interview: Interviewing is a technique that is primarily used to gain an understanding of the underlying reasons and motivations for people's attitudes, preferences or behavior. Interviews can be undertaken on a personal one-to-one basis or in a group.

Experimental Method: There are number of experimental designs that are used in carrying out and experiment. However, Market researchers have used 4 experimental designs most frequently. These are

CRD - Completely Randomized Design

RBD - Randomized Block Design

LSD - Latin Square Design

FD - Factorial Designs

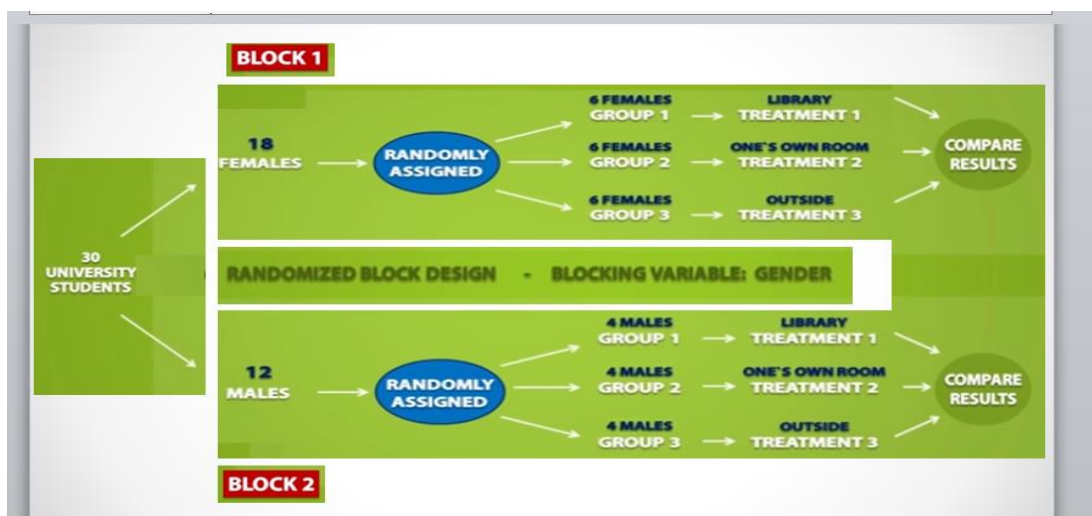
CRD: A completely randomized design (CRD) is one where the treatments are assigned completely at random so that each experimental unit has the same chance of receiving any one treatment.

CRD is appropriate only for experiments with homogeneous experimental units.

Example:



RBD - The term **Randomized Block Design** has originated from agricultural research. In this design several treatments of variables are applied to different blocks of land to ascertain their effect on the yield of the crop. Blocks are formed in such a manner that each block contains as many plots as a number of treatments so that one plot from each is selected at random for each treatment. The production of each plot is measured after the treatment is given. These data are then interpreted and inferences are drawn by using the analysis of Variance Technique so as to know the effect of various treatments like different dozes of fertilizers, different types of irrigation etc.



LSD - Latin Square Design - A Latin square is one of the experimental designs which has a balanced two way classification scheme say for example - 4 X 4 arrangement. In this scheme each letter from A to D occurs only once in each row and also only once in each column. The balance arrangement, it may be noted that, will not get disturbed if any row gets changed with the other.

Definition:

A Latin square is a *square array* of objects (letters A, B, C,...) such that *each object appears once and only once* in each row and each column.

Example :

A, B, C, D

Suppose we choose the following Latin square:

A	B	C	D
B	C	D	A
C	D	A	B
D	A	B	C

Another choice of a Latin square of order 4 is

C	B	A	D
B	C	D	A
A	D	C	B
D	A	B	C

Selected Latin Squares can be in these forms:

3 × 3		
A	B	C
B	C	A
C	A	B

4 × 4			
A	B	C	D
B	C	D	A
C	D	A	B
D	A	B	C

5 × 5				
A	B	C	D	E
B	C	D	E	A
C	D	E	A	B
D	E	A	B	C
E	A	B	C	D

The balance arrangement achieved in a Latin Square is its main strength. In this design, the comparisons among treatments will be free from both differences between rows and columns. Thus the magnitude of error will be smaller than any other design.

In LSD, You have Three Factors:

- 1) Rows
- 2) Columns
- 3) Treatments (letters A, B, C....)

The number of Treatments = No. Of Rows = No. Of column = n

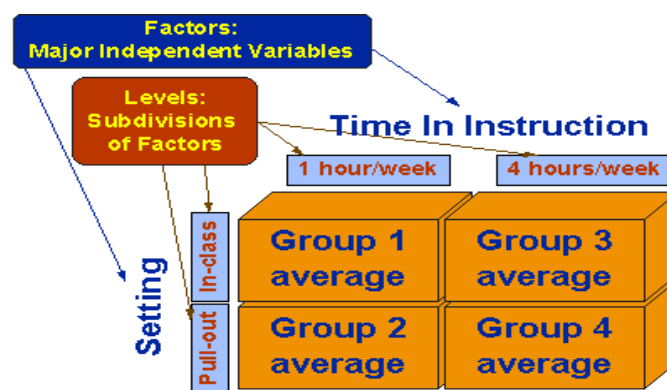
The row-column treatments are represented by cells in a $n \times n$ array.

The treatments are assigned to row-column combinations using a Latin-Square arrangement.

FD - Factorial Designs - This design allows the experimenter to test two or more variables simultaneously. It also measures interaction effects of the variables and analyzes the impacts of each of the variables. In a true experiment, randomization is essential so that the experimenter can infer cause and effect without any bias.

A experiment which involves multiple independent variables is known as a factorial design.

A factor is a major independent variable. In this example we have two factors: time in instruction and setting. A level is a subdivision of a factor. In this example, time in instruction has two levels and setting has two levels.



For example, suppose a botanist wants to understand the effects of sunlight (low vs. high) and watering frequency (daily vs. weekly) on the growth of a certain species of plant.

		Watering Frequency	
		Daily	Weekly
Sunlight	Low	Plant Growth	Plant Growth
	High	Plant Growth	Plant Growth

This is an example of a 2x2 factorial design because there are two independent variables, each with two levels:

Independent variable #1: Sunlight (Levels: Low, High)

Independent variable #2: Watering Frequency (Levels: Daily, Weekly)

2. Secondary sources of Data:

While secondary sources means data collected by someone else earlier. Secondary data are the data collected by a party not related to the

research study but collected these data for some other purpose and at different time in the past.

If the researcher uses these data then these become secondary data for the current users. Sources of secondary data are government publications websites, books, journal articles, internal records.

1. Internal Sources

2. External Sources

Internal Sources –These are within the organization. **External Sources** - These are outside the organization

➤ **Internal Sources:**

If available, internal secondary data may be obtained with less time, effort and money than the external secondary data. In addition, they may also be more pertinent to the situation at hand since they are from within the organization. The internal sources include

Accounting resources- This gives so much information which can be used by the marketing researcher. They give information about internal factors.

Sales Force Report- It gives information about the sale of a product. The information provided is of outside the organization.

Internal Experts- These are people who are heading the various departments. They can give an idea of how a particular thing is working

Miscellaneous Reports- These are what information you are getting from operational reports. If the data available within the organization are unsuitable or inadequate, the marketer should extend the search to external secondary data sources.

➤ **External Sources of Data:**

External Sources are sources which are outside the company in a larger environment. Collection of external data is more difficult because the data have much greater variety and the sources are much more numerous.

Government Publications- Government sources provide an extremely rich pool of data for the researchers. In addition, many of these data are available free of cost on internet websites.

There are number of government agencies generating data.

These are:

Registrar General of India- It is an office which generates demographic data. It includes details of gender, age, occupation etc.

Central Statistical Organization- This organization publishes the national accounts statistics. It contains estimates of national income for several years, growth rate, and rate of major economic activities. Annual survey of Industries is also published by the CSO. It gives information about the total number of workers employed, production units, material used and value added by the manufacturer.

Director General of Commercial Intelligence- This office operates from Kolkata. It gives information about foreign trade i.e. import and export. These figures are provided region-wise and country-wise.

Ministry of Commerce and Industries- This ministry through the office of economic advisor provides information on wholesale price index. These indices may be related to a number of sectors like food, fuel, power, food grains etc. It also generates All India Consumer Price Index numbers for industrial workers, urban, non manual employees and cultural labourers.

Planning Commission- It provides the basic statistics of Indian Economy. Reserve Bank of India- This provides information on Banking Savings and investment. RBI also prepares currency and finance reports.

Labour Bureau- It provides information on skilled, unskilled, white collared jobs National Sample Survey- This is done by the Ministry of Planning and it provides social, economic, demographic, industrial and agricultural statistics.

Department of Economic Affairs- It conducts economic survey and it also generates information on income, consumption, expenditure, investment, savings and foreign trade.

State Statistical Abstract- This gives information on various types of activities related to the state like - commercial activities, education, occupation etc.

Non-Government Publications- These includes publications of various industrial and trade associations, such as The Indian Cotton Mill Association Various chambers of commerce.

Understand various sources of Data like Sensors/signal/GPS etc:

Sensor data:

- Sensor data is the output of a device that detects and responds to some type of input from the physical environment. The output may be used to provide information or input to another system or to guide a process.

- Here are a few examples of sensors, just to give an idea of the number and diversity of their applications:
- ❖ **A photosensor** detects the presence of visible light, infrared transmission (IR) and/or ultraviolet (UV) energy.
 - ❖ **Smart grid sensors** can provide real-time data about grid conditions, detecting outages, faults and load and triggering alarms.
 - ❖ **Wireless sensor networks** combine specialized transducers with a communications infrastructure for monitoring and recording conditions at diverse locations. Commonly monitored parameters include temperature, humidity, pressure, wind direction and speed, illumination intensity, vibration intensity, sound intensity, powerline voltage, chemical concentrations, pollutant levels and vital body functions.

Signal:

The simplest form of signal is a direct current (DC) that is switched on and off; this is the principle by which the early telegraph worked. More complex signals consist of an alternating-current (AC) or electromagnetic carrier that contains one or more data streams.

Data must be transformed into electromagnetic signals prior to transmission across a network. Data and signals can be either analog or digital. A signal is periodic if it consists of a continuously repeating pattern.

Global Positioning System (GPS):

The Global Positioning System (GPS) is a space based navigation system that provides location and time information in all weather conditions, anywhere on or near the Earth where there is an unobstructed line of sight to four or more GPS satellites. The system provides critical capabilities to military, civil, and commercial users around the world. The United States government created the system, maintains it, and makes it freely accessible to anyone with a GPS receiver.

Quality of Data:

Data quality is the ability of your data to serve its intended purpose based on factors such as accuracy, completeness, consistency, reliability and these factors that play a huge role in determining data quality.

Accuracy:

Erroneous values that deviate from the expected. The causes for inaccurate data can be various, which include:

- Human/computer errors during data entry and transmission
- Users deliberately submitting incorrect values (called disguised missing data)
- Incorrect formats for input fields
- Duplication of training examples

Completeness:

Lacking attribute/feature values or values of interest. The dataset might be incomplete due to:

- Unavailability of data
- Deletion of inconsistent data
- Deletion of data deemed irrelevant initially

Consistency: Inconsistent means data source containing discrepancies between different data items. Some attributes representing a given concept may have different names in different databases, causing inconsistencies and redundancies. Naming inconsistencies may also occur for attribute values.

Reliability: Reliability means that data are reasonably complete and accurate, meet the intended purposes, and are not subject to inappropriate alteration.

Some other features that also affect the data quality include **timeliness** (the data is incomplete until all relevant information is submitted after certain time periods), **believability** (how much the data is trusted by the user) and **interpretability** (how easily the data is understood by all stakeholders).

To ensure high quality data, it's crucial to preprocess it. To make the process easier, data preprocessing is divided into four stages: data cleaning, data integration, data reduction, and data transformation.

Data Quality is also effected by

- Outliers
- Missing Values
- Noisy
- Duplicate Values

Outliers:

Outliers are extreme values that deviate from other observations on data, they may indicate a variability in a measurement, experimental errors or a novelty. It

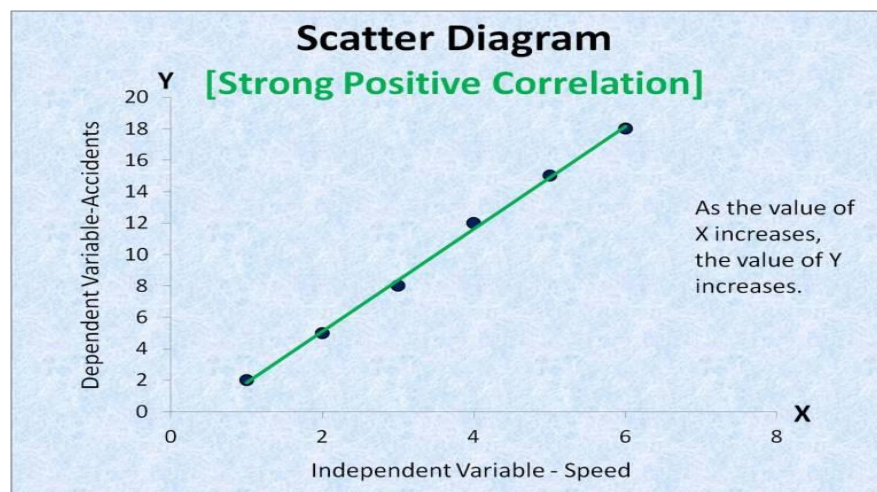
is a point or an observation that deviates significantly from the other observations.

Outlier detection from graphical representation:

- **Scatter plot and**
- **Box plot**

Scatter plot:

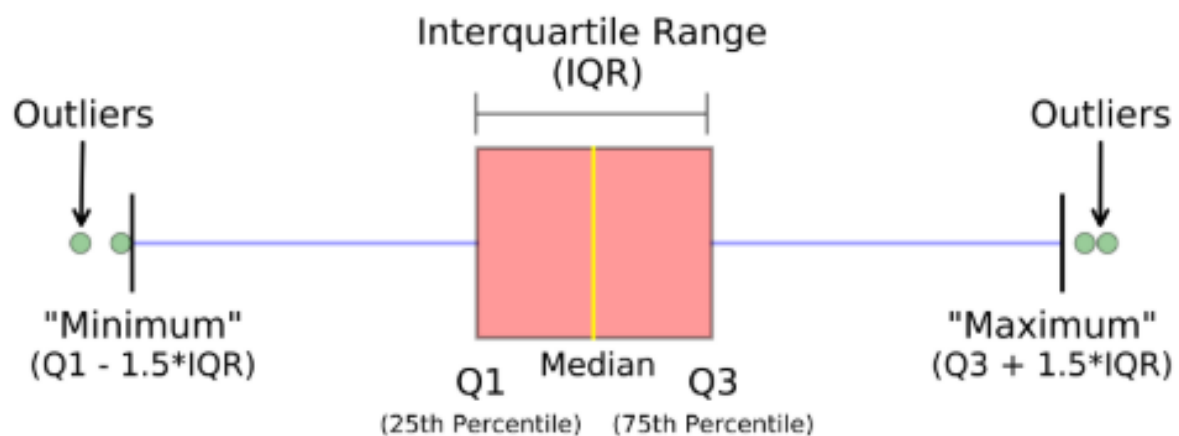
Scatter plots are used to plot data points on a horizontal and a vertical axis in the attempt to show how much one variable is affected by another. A **scatter plot** uses dots to represent values for two different numeric variables.



Box plot:

A **boxplot** is a standardized way of displaying the distribution of data based on a **five number summary**

- Minimum
- First quartile (Q1),
- Median,
- Third quartile (Q3), and
- Maximum").



Most common causes of outliers on a data set:

- Data entry errors (human errors)
- Measurement errors (instrument errors)
- Experimental errors (data extraction or experiment planning/executing errors)
- Intentional (dummy outliers made to test detection methods)
- Data processing errors (data manipulation or data set unintended mutations)
- Sampling errors (extracting or mixing data from wrong or various sources)
- Natural (not an error, novelties in data)

How to remove Outliers?

Most of the ways to deal with outliers are similar to the methods of missing values like deleting observations, transforming them, binning them, treat them as a separate group, imputing values and other statistical methods. Here, we will discuss the common techniques used to deal with outliers:

Deleting observations: We delete outlier values if it is due to data entry error, data processing error or outlier observations are very small in numbers. We can also use trimming at both ends to remove outliers.

Transforming and binning values: Transforming variables can also eliminate outliers. Natural log of a value reduces the variation caused by extreme values. Binning is also a form of variable transformation. Decision Tree algorithm allows to deal with outliers well due to binning of variable. We can also use the process of assigning weights to different observations.

Imputing: Like imputation of missing values, we can also impute outliers. We can use mean, median, mode imputation methods. Before imputing values, we should analyse if it is natural outlier or artificial. If it is artificial, we can go with imputing values. We can also use statistical model to predict values of outlier observation and after that we can impute it with predicted values.

Missing data:

Missing data in the training data set can reduce the power / fit of a model or can lead to a biased model because we have not analysed the behavior and relationship with other variables correctly. It can lead to wrong prediction or classification.

Why my data has missing values?

We looked at the importance of treatment of missing values in a dataset. Now, let's identify the reasons for occurrence of these missing values. They may occur at two stages:

1. Data Extraction: It is possible that there are problems with extraction process. In such cases, we should double-check for correct data with data guardians. Some hashing procedures can also be used to make sure data extraction is correct. Errors at data extraction stage are typically easy to find and can be corrected easily as well.

2. Data collection: These errors occur at time of data collection and are harder to correct. They can be categorized in four types:

- **Missing completely at random:** This is a case when the probability of missing variable is same for all observations. For example: respondents of data collection process decide that they will declare their earning after tossing a fair coin. If an head occurs, respondent declares his / her earnings & vice versa. Here each observation has equal chance of missing value.
- **Missing at random:** This is a case when variable is missing at random and missing ratio varies for different values / level of other input variables. For example: We are collecting data for age and female has higher missing value compare to male.
- **Missing that depends on unobserved predictors:** This is a case when the missing values are not random and are related to the unobserved input variable. For example: In a medical study, if a particular diagnostic causes discomfort, then there is higher chance of drop out from the study. This missing value is not at random unless we have included "discomfort" as an input variable for all patients.
- **Missing that depends on the missing value itself:** This is a case when the probability of missing value is directly correlated with missing value itself. For example: People with higher or lower income are likely to provide non-response to their earning.

Which are the methods to treat missing values?

1. Deletion: It is of two types: **List Wise Deletion** and **Pair Wise Deletion**.

- In **list wise deletion**, we delete observations where any of the variable is missing. Simplicity is one of the major advantage of this method, but this method reduces the power of model because it reduces the sample size.
- In **pair wise deletion**, we perform analysis with all cases in which the variables of interest are present. Advantage of this method is, it keeps as many cases available for analysis. One of the disadvantage of this method, it uses different sample size for different variables.

List wise deletion

Gender	Manpower	Sales
M	25	343
F	.	280
M	33	332
M	.	272
F	25	.
M	29	326
.	26	259
M	32	297

Pair wise deletion

Gender	Manpower	Sales
M	25	343
F	.	280
M	33	332
M	.	272
F	25	.
M	29	326
.	26	259
M	32	297

- Deletion methods are used when the nature of missing data is “**Missing completely at random**” else non random missing values can bias the model output.

2. Mean/ Mode/ Median Imputation: Imputation is a method to fill in the missing values with estimated ones. The objective is to employ known relationships that can be identified in the valid values of the data set to assist in estimating the missing values. Mean / Mode / Median imputation is one of the most frequently used methods. It consists of replacing the missing data for a given attribute by the mean or median (quantitative attribute) or mode (qualitative attribute) of all known values of that variable.

It can be of two types:

- **Generalized Imputation:** In this case, we calculate the mean or median for all non missing values of that variable then replace missing value with mean or median. Like in above table, variable “Manpower” is missing so we take average of all non missing values of “Manpower” (28.33) and then replace missing value with it.
- **Similar case Imputation:** In this case, we calculate average for gender “Male” (29.75) and “Female” (25) individually of non missing values then replace the missing value based on gender. For “Male“, we will replace missing values of manpower with 29.75 and for “Female” with 25.

Noisy Data:

Noisy data is a meaningless data that can't be interpreted by machines. It can be generated due to faulty data collection, data entry errors etc. It can be handled in following ways:

Binning Method:

This method works on sorted data in order to smooth it. The whole data is divided into segments of equal size and then various methods are performed to complete the task. Each segmented is handled separately. One can replace all data in a segment by its mean or boundary values can be used to complete the task.

- **Smoothing by bin means:** In smoothing by bin means, each value in a bin is replaced by the mean value of the bin.
- **Smoothing by bin median:** In this method each bin value is replaced by its bin median value.
- **Smoothing by bin boundary:** In smoothing by bin boundaries, the minimum and maximum values in a given bin are identified as the bin boundaries. Each bin value is then replaced by the closest boundary value.

Regression: Here data can be made smooth by fitting it to a regression function. The regression used may be linear (having one independent variable) or multiple (having multiple independent variables).

Clustering: This approach groups the similar data in a cluster. The outliers may be undetected or it will fall outside the clusters.

Incorrect attribute values may due to

- faulty data collection instruments
- data entry problems data entry problems
- data transmission problems
- technology limitation
- inconsistency in naming convention

Duplicate values: A dataset may include data objects which are duplicates of one another. It may happen when say the same person submits a form more than once. The term deduplication is often used to refer to the process of dealing with duplicates. In most cases, the duplicates are removed so as to not give that particular data object an advantage or bias, when running machine learning algorithms.

Redundant data occurs while we merge data from multiple databases. If the redundant data is not removed incorrect results will be obtained during data analysis. Redundant data occurs due to the following reasons.

- **Object identification:** The same attribute or object may have different names in different databases
- **Derivable data:** One attribute may be a “derived” attribute in another table, e.g., annual revenue

Redundant attributes may be able to be detected by *correlation analysis* Careful integration of the data from multiple sources may help reduce/avoid redundancies and inconsistencies and improve mining speed and quality

Data Pre-processing:

Data preprocessing is a data mining technique that involves transforming raw data into an understandable format. Real-world data is often incomplete, inconsistent, and/or lacking in certain behaviors or trends, and is likely to contain many errors.

Major Tasks in Data Preprocessing are

- Data Cleaning
- Data Integration
- Data Transformation
- Data Reduction

1. Data Cleaning: Data is cleansed through processes such as filling in missing values, smoothing the noisy data, or resolving the inconsistencies in the data.

Data cleaning tasks

- Fill in missing values
- Identify outliers and smooth out noisy data
- Correct inconsistent data
- Resolve redundancy caused by data integration

Incomplete: lacking attribute values, lacking certain attributes of interest, or containing only aggregate data

e.g., occupation=“ ”

Noisy: containing errors or outlier value that deviate from the expected.

e.g., Salary=“-10”

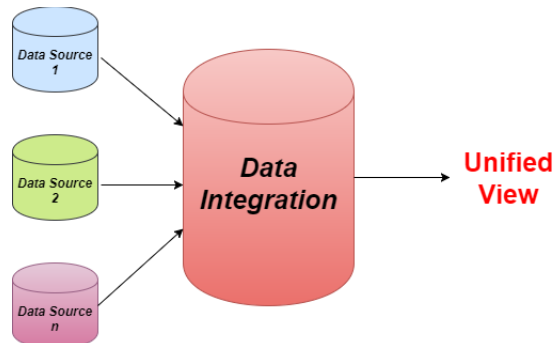
Inconsistent: containing discrepancies in codes or names

e.g., Age=“42” Birthday=“03/07/1997”

e.g., Was rating “1,2,3”, now rating “A, B, C”

e.g., discrepancy between duplicate records

2. Data Integration: Data with different representations are put together and conflicts within the data are resolved. Integration of multiple databases, data cubes, or files.



There are mainly 2 major approaches for data integration – one is “Tight coupling approach” and another is “Loose coupling approach”.

Tight Coupling:

Here, a data warehouse is treated as an information retrieval component. In this coupling, data is combined from different sources into a single physical location through the process of ETL – Extraction, Transformation and Loading.

Loose Coupling:

Here, an interface is provided that takes the query from the user, transforms it in a way the source database can understand and then sends the query directly to the source databases to obtain the result.

Issues in Data Integration:

There are three issues to consider during data integration: Schema Integration, Redundancy Detection, and resolution of data value conflicts. These are explained in brief below.

- **Schema Integration:** Integrate metadata from different sources. The real-world entities from multiple sources are referred to as the entity identification problem.
- **Redundancy:** An attribute may be redundant if it can be derived or obtained from another attribute or set of attributes. Inconsistencies in attributes can also cause redundancies in the resulting data set. Some redundancies can be detected by correlation analysis.
- **Detection and resolution of data value conflicts:** This is the third critical issue in data integration. Attribute values from different sources may differ

for the same real-world entity. An attribute in one system may be recorded at a lower level of abstraction than the “same” attribute in another.

3. Data Transformation: This step is taken in order to transform the data in appropriate forms suitable for mining process. This involves following ways.

- **Normalization:** It is done in order to scale the data values in a specified range (-1.0 to 1.0 or 0.0 to 1.0).

Min-Max Normalization:

This transforms the original data linearly. Suppose that $\min_F$ is the minima and $\max_F$ is the maxima of an attribute, F

We Have the Formula:

$$v' = \frac{v - \min_F}{\max_F - \min_F} (\text{new_max}_F - \text{new_min}_F) + \text{new_min}_F,$$

Where v is the value you want to plot in the new range. v' is the new value you get after normalizing the old value.

- **Attribute Selection:** In this strategy, new attributes are constructed from the given set of attributes to help the mining process.
- **Discretization:** This is done to replace the raw values of numeric attribute by interval levels or conceptual levels.
- **Concept Hierarchy Generation:** Here attributes are converted from level to higher level in hierarchy. For Example-The attribute “city” can be converted to “country”.

4. Data Reduction: Since data mining is a technique that is used to handle huge amount of data. While working with huge volume of data, analysis became harder in such cases. In order to get rid of this, we use data reduction technique. It aims to increase the storage efficiency and reduce data storage and analysis costs.

The various steps to data reduction are:

- **Data Cube Aggregation:** Aggregation operation is applied to data for the construction of the data cube.
- **Attribute Subset Selection:** The highly relevant attributes should be used, rest all can be discarded. For performing attribute selection, one can use level of significance and p-value of the attribute. The attribute having p-value greater than significance level can be discarded.

➤ **Data compression**

It reduces the size of the files using different encoding mechanisms (Huffman Encoding & run-length Encoding). We can divide it into two types based on their compression techniques.

Lossless Compression:

Encoding techniques (Run Length Encoding) allows a simple and minimal data size reduction. Lossless data compression uses algorithms to restore the precise original data from the compressed data.

Lossy Compression:

Methods such as Discrete Wavelet transform technique, PCA (principal component analysis) are examples of this compression. In lossy-data compression, the decompressed data may differ to the original data but are useful enough to retrieve information from them.

- **Numerosity Reduction:** This enables to store the model of data instead of whole data, for example: Regression Models.
- **Dimensionality Reduction:** This reduces the size of data by encoding mechanisms. It can be lossy or lossless. If after reconstruction from compressed data, original data can be retrieved, such reduction are called lossless reduction else it is called lossy reduction. The two effective methods of dimensionality reduction are: Wavelet transforms and PCA (Principal Component Analysis).

5. Data Discretization: Involves the reduction of a number of values of a continuous attribute by dividing the range of attribute intervals. Data discretization refers to a method of converting a huge number of data values into smaller ones so that the evaluation and management of data become easy. In other words, data discretization is a method of converting attributes values of continuous data into a finite set of intervals with minimum data loss.

Suppose we have an attribute of Age with the given values

Age	1.5.9.4.7.11.14.17.13.18. 19.31.33.36.42.44.46.70.74.78.77
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Attribute	Age	Age	Age	Age
	1,5,4,9,7	11,14,17,13,18,19	31,33,36,42,44,46	70,74,77,78
After Discretization	Child	Young	Mature	Old

Important Questions

1. Discuss about Data Management? How to manage the data for analysis?
2. How to design Data Architecture? What are the factors that influences the data architecture?
3. Define Primary Data sources and also explain about the types of primary data sources?
4. What is Data Pre-processing? Discuss about various steps to pre-process the data?
5. Explain about Missing values and how to eliminate missing values?
6. Define Quality of data and discuss various factors that affect the Quality of data?
7. Discuss about various sources of data?
8. What is Noisy data? Explain about methods to handle noisy data?
9. Discuss about different methods in experimental data sources (CRD, RBD, LSD and FA)?
10. Differentiate internal and external secondary data sources with examples?
11. What is data transformation in data preprocessing and discuss about different normalization techniques?
12. Discuss the process of handling duplicate values in organizational data. Briefly describe various sources of data like sensors, signals GPS in data management?